

Case Study



Sanofi

How they focused on developing
Sustainable Innovation & Patents using TRIZ
and increased sales from
\$1.5 billion to \$6 billion in 6 years

Sustainability means you are continuously looking at innovation and improvement

Mick Bremans, Former CEO
Ecover.

Excellent firms don't believe in excellent, only in constant improvement and constant change

Tom Peters, Business
Author.

Using TRIZ to deliver Sustainable Innovation

Sanofi is a French multinational pharmaceutical company with headquarters in Paris. Sanofi engages in the research and development, manufacturing and marketing of pharmaceutical drugs and covers seven major areas: cardiovascular, central nervous system, diabetes, internal medicine, oncology, thrombosis and vaccines.

In 2001 Sanofi was growing concerned. They ranked third in Diabetes – out of the three largest companies involved with Insulin products, but they had a poor range of devices. They wanted to be the first in Diabetes and have a more competitive range of good quality, usable Insulin devices. However, they were only submitting **6 Patents per year** as they battled reliability, usability problems, product recalls and supply dramas.

Clearly, they needed to overhaul their idea generation process and focus on Sustainable Innovation.

What is Sustainable Innovation?

Sustainable Innovation means that you are continuously looking for better solutions, no product is ever finished. The key point of sustainable innovation is **managing your knowledge to innovate**. Sustainable Innovation is important, and we only need to look at the FTSE100 and Fortune 500 lists to see why. Since 1984 only two companies have been a constant in the FTSE100 (Rothschild and Reuters) and in the past 50 years, 88% of companies in the Fortune 500 have disappeared.

Sanofi challenged Oxford Creativity to review their idea generation process and increase the number of ideas consistently generated to help fuel their sustainable improvement.

Oxford Creativity and Sanofi

Oxford Creativity joined a fantastic in-house team at Sanofi which had been hand-picked to focus entirely on improving the idea generation process. The aim was to create a 'War Chest' of Patents to allow Sanofi to continue to innovate sustainably for years to come.

The Guiding Principle used by the team was 'organisations can only cope with four things at a time' – people tend to be the same. We wanted to keep it simple and focussed.

The first thing for Oxford Creativity and the team to do was decide which tools should be used. In order to do this, they asked themselves the following questions:

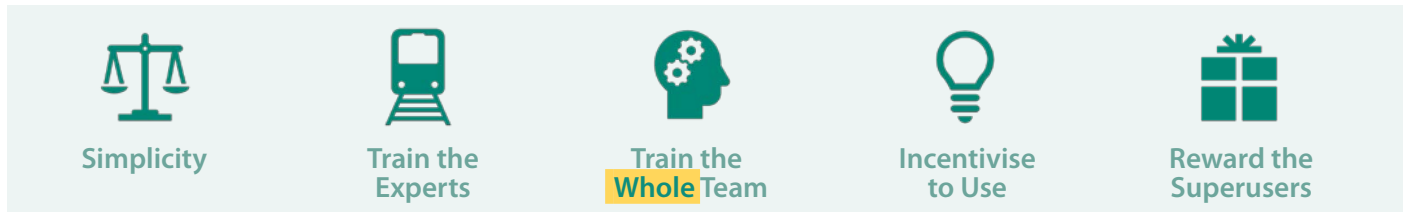
- What were we there for?
- What didn't we know?
- How did we want to work?
- What did we have to deliver?
- How much time did we have?
- What was going to happen to us?
- How much money did we have to spend?

They decided to start with the tools: Good Practice; Understanding; Problem Solving and Working with Others.

Deploying the Tools

Oxford Creativity then had to ensure the tools were well deployed. This started with simplicity, as they ensured the tools were simple and easy to follow. In-house at Sanofi, they then brought in and trained the experts to ensure they understood exactly what they needed to do.

Then Oxford Creativity trained the whole team, including all the assistants, so everyone was on the same page. They incentivised the use of the tools, to ensure uptake and finally they made sure to reward those who used the tools the most – the superusers!



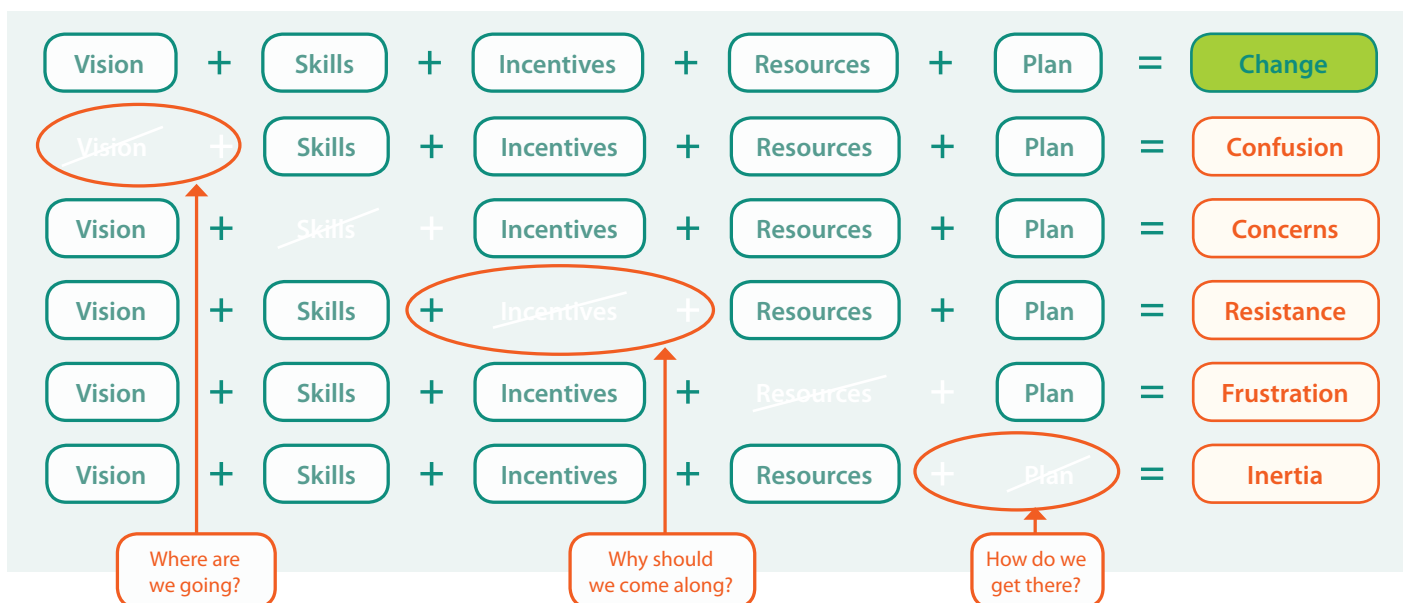
Oxford Creativity and the team recognised how important it was to be open to influence and know who to influence. In reality this meant working out who in the team would be the most receptive to the tools and training them first, to ensure they would be a driving force.

This meant they had a self-belief, willingness to fail, hunger for learning and they were playing for keeps – they were aiming high.

Creating a Strategy

Oxford Creativity and the team knew that Sanofi's Sustainable Innovation initiative needed to have a strategy. Without a strategy an initiative will never be put into place. *Figure 1* illustrates the issues that initiatives come up against without a strategy. If there is no clear vision, then your team won't know where they are going or what they are trying to achieve, and it will lead to confusion. If the strategy doesn't contain incentives, then the team won't know why they should put in the effort.

Figure 1



Finally, the strategy must include a plan, if there is no plan then the team will not know how to put the change in place – and will result in no real change.

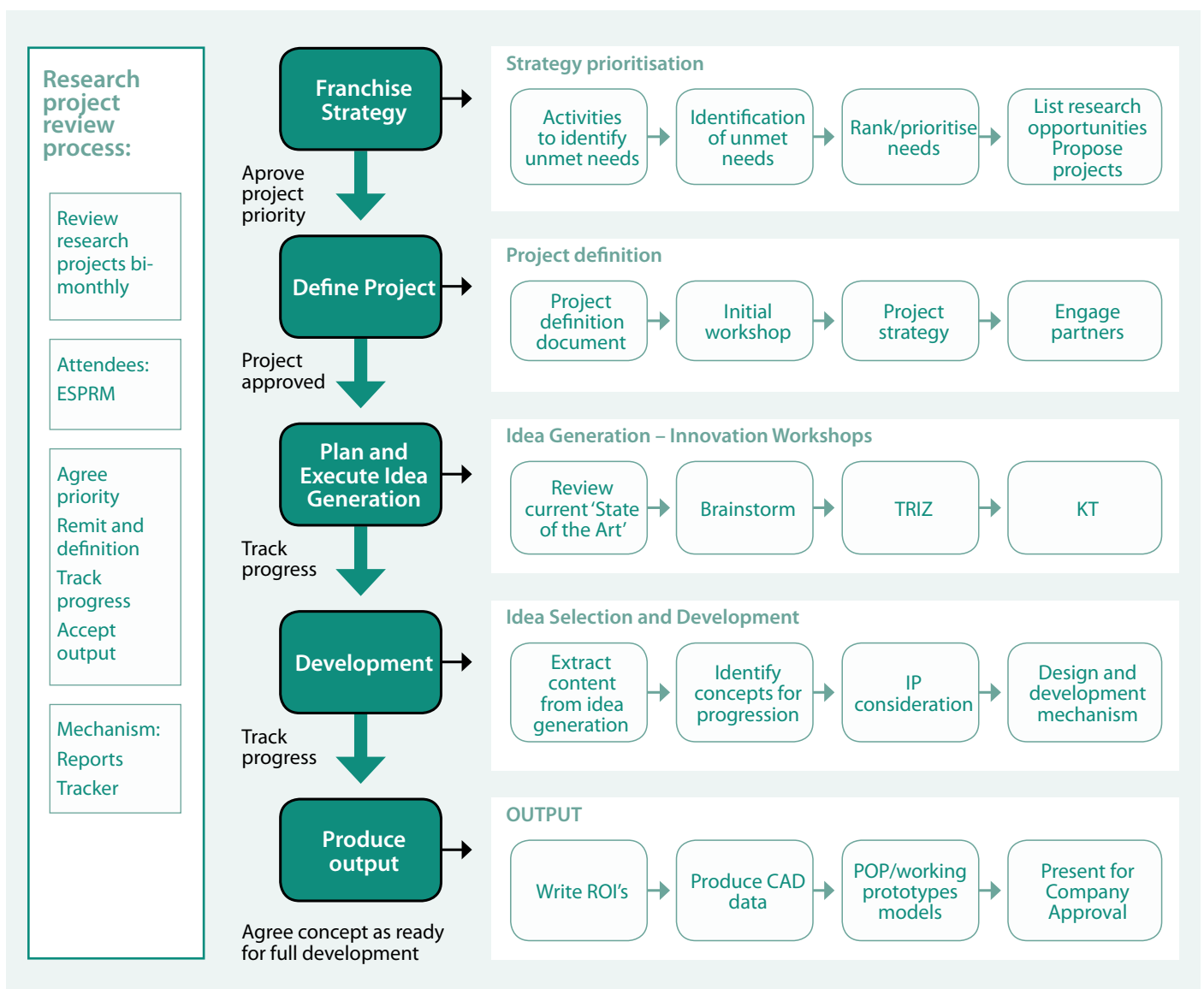
After discussing the best strategy for implementing Sustainable Innovation within Sanofi Oxford Creativity and the team decided to 'Fix the Mix' and change the tools used to focus on: Good Practice, Building Trust, Usability and TRIZ. Certain members of the team were identified as people who would get along very well with TRIZ, so they were taught to use it first, so eventually they could support the rest of the team.

The Brief

Oxford Creativity and the team knew that in order to implement Sustainable Innovation they needed a more specific brief. After some discussion the brief decided on was to generate a Patent 'War Chest'.

They used Sanofi's current idea generation process as a starting point. They identified a few issues with the original process very quickly. Firstly, it had an entirely Pharmaceutical focus, which meant it missed out on other relevant innovations. In

Figure 2 Sanofi's Original Idea Generation Process



addition to this, it developed only to the needs of users; it was **reactive not proactive**, and all the design was done externally so Sanofi had little control over the entire process. The team decided it was important that the new idea generation process would have a high emphasis on quality, and also was effective and efficient, as there were limited resources. This meant Sanofi needed to be able to:

- Set the Standard
- Control the Process
- Have the ability to Stop and Reset the Process
- Build in Continuous Improvement

The New Idea Generation Process

When Oxford Creativity and the team began to devise a new process, they decided it should be a cyclical process that started and ended with Sanofi's company strategy – all ideas must be relevant to their company strategy.

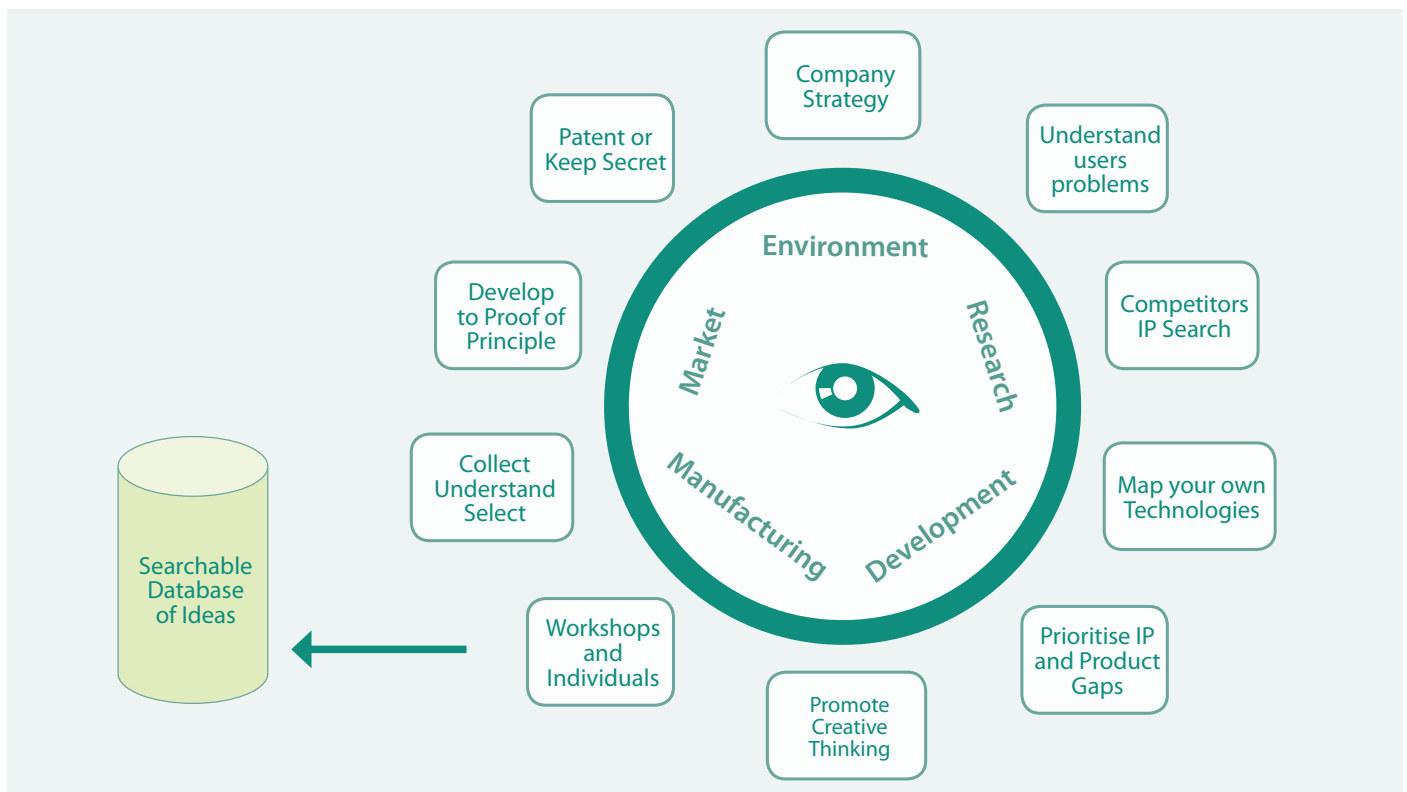


Figure 3 Sanofi's New Idea Generation Process

Another feature of this new process was its focus on Sustainable Innovation, this meant that not every idea should be patented, and thus made public, immediately. The team realised that built into the process should be a decision on whether to patent the idea or keep it secret. Over four years this process has evolved into a highly effective idea generation process.

Company Strategy

It was important that the idea generation process started with the company strategy and that all efforts were aligned to the company strategy. This is one of the most

important parts of the idea generation process so the first thing to do is put a company strategy in place before anything else.



Figure 4 The McKinsey Model

Understanding User Problems

Once the Company Strategy had been understood, Oxford Creativity and the team then had to understand user problems. In order to do this, they had meetings with Key Opinion Leaders, shadowed users and created Job Maps. They also created User Personas so idea generators could really understand what Sanofi's users want and need.

It is important that this was built into the process, as that the users want is variable and thus ever-changing.

Competitors IP Search

Oxford Creativity and the team understood that in order to put Sustainable Innovation into practice, the idea generation process must take into account what Sanofi's competitors are doing. During this part of the idea generation process is about collecting **Competitive Intelligence** – intelligence which adds value. This means seeking out information about competitors by looking into:

- Who
- IP
- Products
- Problems
- Market
- Research

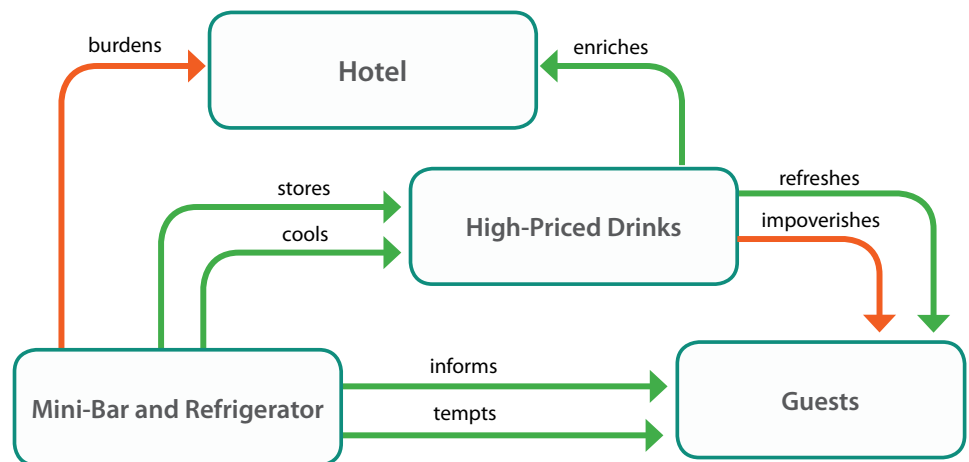
This is where TRIZ had become a valuable tool within Sanofi, as it very quickly helped those working on innovation to identify and understand patterns within the information.

Oxford Creativity and the team at Sanofi realised how important it was that this information would then be shared with everyone. In order to do this, they created a Physical Library of Products which is now constantly being added to and helped the team members to understand the usability of their competitors' products.

“It is wiser to
find out than to
suppose”
Mark Twain

Function Mapping

Once the Competitor IP search was completed the next stage in the idea generation process was Function Mapping. It was important to Oxford Creativity and the team to include this in the processes as it is such a powerful tool.



By mapping the processes from start to finish, those working on innovation are able to see what you are doing for each step and why, this provides remediation in highly regulated industries. If the resources are available, it is best to also map competitor devices.

The Sanofi team quickly realised that function mapping would feed into their Continuous Improvement initiative. By mapping out the existing steps and features in a given process, they could easily see which steps are unnecessary and remove them.

Function Mapping is now a vital step in Sanofi's idea generation process. They include Market feedback and chunk the functioning to get a really good idea of what products they already provide.



Prioritise Product Gaps

Oxford Creativity and the team at Sanofi understood the power of being able to answer the question "Why do we do that?"

By building 'Prioritising IP and Products Gaps' into the idea generation process, they were ensuring Sanofi would always be able to answer that question. This was driven by strategy, it combined information from function mapping, the manufacturing process, customers and marketing organisations to create Sustainable Innovation.

Promote Creative Thinking

Creative Thinking is Key in Sustainable Innovation, so it was important that Sanofi created an environment where Creative Thinking thrived. In order to do this Oxford Creativity and the Sanofi team put in place processes which ensured creativity was rewarded. They recognised how important follow-up was, so that an idea could be tracked from the original post-it-note to its eventual implementation. Having a process which is transparent to those contributing, by opening up the relevant database and making it accessible but not changeable, helped foster creative thinking within Sanofi.

Workshops and Individuals

Workshops are one of the most important aspects of Sustainable Innovation, without new ideas you cannot innovate, and workshops are one of the best ways to come up with new ideas.

However, it is important these workshops have the right people doing the right things. Oxford Creativity and the rest of the team realised that to ensure this happened workshops must be guided by Company Strategy, start with setting realistic tangible objectives which are time achievable and end by setting relevant homework. This could be Patent searches or feedback investigation, but it must be defined with the group.

Collect, Understand, Select

One of the key tools within Oxford TRIZ is being able to collect and understand the ideas generated in the workshops. In order to do this, Oxford Creativity make smart use of the humble post-it-note.

Sanofi adopted the same process to ensure no ideas were lost. After all workshops each post-it-note was identified by when it was created and who created it.

This enabled Sanofi to generate and continually update a time-based database which contained every idea, when it was thought of and who by. Helping Sanofi reward its team members when they implement one of their ideas and encourage more great ideas!

Proof of Principle

Oxford Creativity recognised that to put Sustainable Innovation in place within Sanofi the ideas which have been generated must be more than just good ideas. They need to work, as they were intended to.

This is why the 'Proof of Principle' step was added in to the idea generation process. The idea must be **reduced to practice** – proven that it will work as it is intended to do, before the decision to patent or not is made.

Patent or Keep Secret?

Once the Principles have been proven to work, a decision must be made whether to file an invention disclosure, and later a patent, or to keep the idea a secret.

Oxford Creativity wanted to ensure the decision-making process was structured and opened out to a wider group within the company. Since these patents cost money, it was important to have a degree of control in this process and ensure that the most important ideas get put forward to patent first. Once the decision had been made it is acted upon immediately to ensure no time was wasted.

Does it Work?

In 2001 Sanofi had a tactical based IP Strategy which only developed around six patents a year. With the Strategy and Idea Generation process put in place by Oxford Creativity and the team at Sanofi, this increased to 276 in 2007.

This created a 'War Chest' of Patents which have stood the company in good stead for the last twelve years and will continue to do so for years to come. By putting in place



Oxford Creativity's idea generation system, Sanofi have become the go-to company for usable insulin devices and had the best in class insulin and device from 2006-2016.

They quadrupled their sales from \$1.5 billion to \$6 billion and have robust products with low complaint rates, user pride and a solid supply chain.

Oxford Creativity working with the Sanofi in-house team of specialists implemented Sustainable Innovation within Sanofi using the power of Process, Individuals and TRIZ.

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